DJS ISACA

PRESENTS

D3COY 1.0

JOIN THE HUNT

PS 01: AI-BASED NATIONAL CYBER THREAT EARLY WARNING SYSTEM

PROBLEM STATEMENT

Cyber threats are evolving rapidly, with attackers exploiting newly discovered vulnerabilities, launching phishing campaigns, distributing malware, and targeting digital infrastructure. Security teams often receive threat information from multiple sources, making it difficult to identify emerging threats, correlate related incidents, and determine which systems require immediate attention.

Traditional security approaches often react after suspicious activity has already occurred. There is a need for an intelligent, proactive system that can analyse available cyber threat intelligence, identify emerging attack patterns, and generate early warnings before threats develop into major security incidents.

CHALLENGE

Develop an AI-powered Cyber Threat Early Warning System that collects and analyses threat intelligence from multiple sources to identify suspicious patterns, emerging attack campaigns, and potentially vulnerable systems.

The system should transform raw threat information into actionable intelligence, helping security analysts understand what is happening, which assets may be at risk, and what preventive measures should be prioritised.

EXPECTED SOLUTION

The proposed solution should demonstrate how AI can help security teams move from reactive incident response towards proactive threat identification and prevention.

A working prototype could include a dashboard showing incoming threat intelligence, an AI-generated threat summary, a prioritised risk score, and an early warning explaining why a particular threat deserves attention.

DJS ISACA

PRESENTS

D3COY 1.0

JOIN THE HUNT

KEY OBJECTIVES

- Threat Intelligence Aggregation: Collect and consolidate threat information from trusted feeds, public advisories, vulnerability databases, and simulated security logs.
- AI-Based Threat Analysis: Apply machine learning or natural language processing to identify suspicious indicators, recurring attack patterns, and relationships between reported threats.
- Early Warning Generation: Identify potentially emerging threats and generate timely alerts based on threat severity, confidence, and relevance.
- Risk Prioritisation: Rank threats according to factors such as exploitability, asset criticality, potential impact, and available evidence.
- Actionable Recommendations: Provide suggested mitigation measures, such as patching vulnerable systems, blocking known malicious indicators, or investigating suspicious activity.
- Threat Intelligence Dashboard: Display active threats, affected technologies

Example use case: A newly published vulnerability advisory is correlated with simulated asset inventory data showing that an organisation uses the affected software. The system identifies the potential exposure, assigns a risk level, and generates an alert recommending investigation and remediation.

EXPECTED DELIVERABLES

- A functional prototype of the AI-based early warning system.
- A dashboard displaying detected threats, risk levels, and alerts.
- A demonstration using public threat intelligence or a controlled, simulated dataset.
- An explanation of the AI model, threat scoring approach, and proposed mitigation workflow.

DJS ISACA

PRESENTS

D3COY 1.0

JOIN THE HUNT

PS 02: AI-BASED DEEPFAKE-ENABLED SOCIAL ENGINEERING DETECTION

PROBLEM STATEMENT

Generative AI has made it easier to create convincing fake audio, video, images, and messages that imitate real individuals or trusted organisations. These capabilities can be misused in phishing, executive impersonation, fraudulent payment requests, identity deception, and other social engineering attacks.
For example, an attacker could impersonate a senior executive through a voice call, send an AI-generated video message requesting an urgent fund transfer, or combine a spoofed identity with a convincing phishing message to manipulate an employee into disclosing confidential information.

Traditional detection methods that rely on spelling mistakes, suspicious wording, or obvious visual inconsistencies are becoming less dependable as synthetic content improves. India's CERT-In has addressed deepfake-related threats, and Gartner's September 2026 survey reported that 41% of surveyed CISOs had encountered at least one social engineering incident involving a deepfake during an employee audio call in the preceding 12 months.

CHALLENGE

Develop an AI-powered system that detects and assesses potential deepfake-enabled social engineering attempts by analysing digital content and the context in which it is presented.

The system should help users identify suspicious communications, understand the indicators that contributed to a risk assessment, and take appropriate verification steps before trusting a message or acting on a request.

EXPECTED SOLUTION

- The solution should provide a practical way to assess potentially fraudulent communications before users respond to them or perform sensitive actions.
- A working prototype could allow users to upload a sample audio or video file, submit a suspicious message, or provide both for analysis. The system would then display relevant detection results, a risk assessment, the indicators identified, and recommended verification steps.

DJS ISACA

PRESENTS

D3COY 1.0

JOIN THE HUNT

KEY OBJECTIVES

- Synthetic Media Detection: Analyse uploaded audio, video, or images for potential signs of manipulation or AI generation using suitable detection models.
- Suspicious Message Analysis: Examine emails, chat messages, or text accompanying media for phishing indicators, urgency-based manipulation, impersonation, and fraudulent requests.
- Multimodal Risk Assessment: Where feasible, combine media-analysis results with textual and contextual indicators to produce a more comprehensive risk assessment.
- Identity and Context Verification: Identify inconsistencies in names, claimed roles, communication context, or requests for sensitive actions. The system may recommend verification through a separately trusted channel.
- Explainable Detection: Highlight suspicious indicators and explain why a communication has been classified as low, medium, or high risk.
- User Alerting and Reporting: Present clear warnings and recommend actions such as independently verifying the sender, reporting the communication, or escalating it to a security team.

EXPECTED DELIVERABLES

- A functional prototype for analysing suspicious messages and/or synthetic media.
- A user interface displaying risk assessments and relevant warning indicators.
- A demonstration using authorised, public, or artificially generated test samples.
- Documentation explaining the detection methodology, model limitations, and recommended response workflow.

DJS ISACA

PRESENTS

D3COY 1.0

JOIN THE HUNT

PS 03: AI-BASED CRITICAL INFRASTRUCTURE CYBER-RISK PREDICTION

PROBLEM STATEMENT

Critical infrastructure forms the backbone of modern society. Power grids, water treatment facilities, transportation networks, telecommunications systems, hospitals, and smart-city infrastructure depend on interconnected digital systems to deliver essential services. As these systems increasingly rely on Internet of Things (IoT) devices, cloud platforms, industrial control systems (ICS), and operational technology (OT), their exposure to cyber threats continues to expand. A vulnerability in one component, a compromised vendor, an exposed device, or an unusual network activity pattern can create risks that extend across interconnected systems.

Unlike conventional IT environments, attacks on critical infrastructure can have consequences beyond data loss, including service interruptions, financial damage, and potential risks to public safety. Identifying vulnerabilities individually is not enough; organisations also need to understand how these vulnerabilities interact, which assets are most critical, and where a cyber incident could have the greatest impact.

India's CERT-In has published cybersecurity guidelines for smart-city infrastructure, addressing areas such as IoT security, continuous monitoring, vulnerability management, supply-chain security, and incident response. Recent international guidance also emphasises the need to manage cybersecurity risks in operational technology while preserving the safety and reliability of essential services.

CHALLENGE

Develop an AI-powered Cyber-Risk Prediction and Prioritisation System that evaluates the security posture of critical infrastructure using simulated infrastructure data, vulnerability information, network activity, and asset criticality.
The system should identify potentially vulnerable components, estimate their associated cyber risk, predict which assets may require urgent attention, and recommend preventive measures before a security incident causes significant disruption.

DJS ISACA

PRESENTS

D3COY 1.0

JOIN THE HUNT

KEY OBJECTIVES

- Infrastructure Asset Profiling: Represent infrastructure assets such as power-grid components, water-treatment systems, hospital networks, IoT devices, and industrial control systems, including their importance and dependencies.
- Vulnerability Assessment: Analyse simulated asset inventories, known vulnerability information, software versions, configuration weaknesses, and exposure indicators to identify potential security gaps.
- AI-Based Risk Prediction: Use machine learning or risk-modelling techniques to estimate the likelihood and potential impact of cyber incidents based on historical or simulated data.
- Risk Prioritization: Rank assets according to vulnerability severity, exposure, operational importance, and potential consequences of compromise.
- Dependency and Impact Analysis: Identify how a compromised component could affect connected assets or essential services.
- Predictive Risk Dashboard: Visualise risk scores, vulnerable assets, changing threat levels, and the components requiring immediate attention.
- Preventive Recommendations: Suggest actions such as patch prioritization, network segmentation, access-control improvements, enhanced monitoring, and incident-response planning.

EXPECTED SOLUTION

- The proposed system should help infrastructure operators transition from reactive vulnerability management to proactive cyber-risk assessment.
- A working prototype could include a simulated infrastructure map, an asset inventory, vulnerability information, AI-generated risk scores, and recommendations for reducing risk.
- The solution should explain why an asset has been classified as high risk and how addressing a particular vulnerability could reduce the overall risk to the infrastructure.

DJS ISACA

PRESENTS

D3COY 1.0

JOIN THE HUNT

PS 04: DECEPTION-DRIVEN THREAT DETECTION AND RESPONSE: A SOC DASHBOARD POWERED BY A HONEYPOT NETWORK

PROBLEM STATEMENT

Security Operations Centers (SOCs) receive thousands of alerts every day from firewalls, endpoints, and system logs, and most of them turn out to be false positives. Analysts spend their time sorting noise, while real attackers can stay inside a network for days before anyone notices. Traditional detection depends on known signatures and suspicious patterns hidden inside normal traffic, so it can rarely say with confidence that an activity is hostile.

Deception technology offers a different signal. A honeypot is a decoy system, such as a fake SSH server, web application, or database, that has no legitimate business use. Any interaction with it is suspicious by definition, which makes it a high-fidelity indicator of compromise. In practice, however, honeypot data often stays in raw log files and never reaches the SOC in a form analysts can act on.

For example, an attacker scans a network, finds a decoy server that looks like an exposed database, logs in with default credentials, and downloads a "customer data" file that is actually fake. Without integration, the SOC sees none of this as a connected attack.

CHALLENGE

Develop a deception-based defence system that deploys honeypots and decoy assets, captures attacker activity in real time, and presents it through a SOC dashboard that helps analysts understand, prioritize, and respond to threats.

The system should help analysts identify who is attacking, what they are doing, how far the attack has progressed, and what action to take, while keeping the real production environment safe.

Example use case: An attacker scans the network and connects to a decoy database. They try default credentials, run commands to list files, and download a decoy "customer data" file. The system identifies the source, captures every command, maps the activity to ATT&CK techniques, checks whether the same source touched any real asset, raises a high-priority alert, and recommends blocking the source, all visible on one screen as a single attack chain.

DJS ISACA

PRESENTS

D3COY 1.0

JOIN THE HUNT

KEY OBJECTIVES

- Honeypot Deployment: Set up one or more realistic decoy services (for example SSH, web login, database, or IoT device) that capture login attempts, commands, and uploaded or downloaded files.
- Real-Time Event Collection: Collect and normalize honeypot interactions into a structured format that a SOC can use, such as a SIEM-style log pipeline.
- Attack Classification and Mapping: Classify activity (scanning, brute force, exploitation, post-compromise commands) and map it to MITRE ATT&CK techniques and kill chain stages.
- Alert Prioritization: Separate opportunistic internet noise from targeted attackers, and assign clear severity levels (low, medium, high, critical) to reduce analyst fatigue.
- Correlation and Attacker Profiling: Link honeypot activity to other telemetry, such as firewall or authentication logs, and build a profile of each attacker showing source, tools, behavior, and attack timeline.
- Automated Response and Threat Intelligence: Recommend or trigger safe actions such as blocking a source, isolating a host, or generating indicators of compromise (IOCs) for sharing.
- Containment and Safety: Make sure the honeypot cannot be used as a pivot point into the production network.

EXPECTED DELIVERABLES

- A functional honeypot setup with at least one realistic decoy service.
- A SOC dashboard displaying live events, prioritized alerts, attacker profiles, and ATT&CK mapping.
- A demonstration using simulated attack traffic generated in an authorised, isolated environment.
- Documentation explaining the deception design, network isolation, detection and prioritization logic, and recommended incident response workflow.